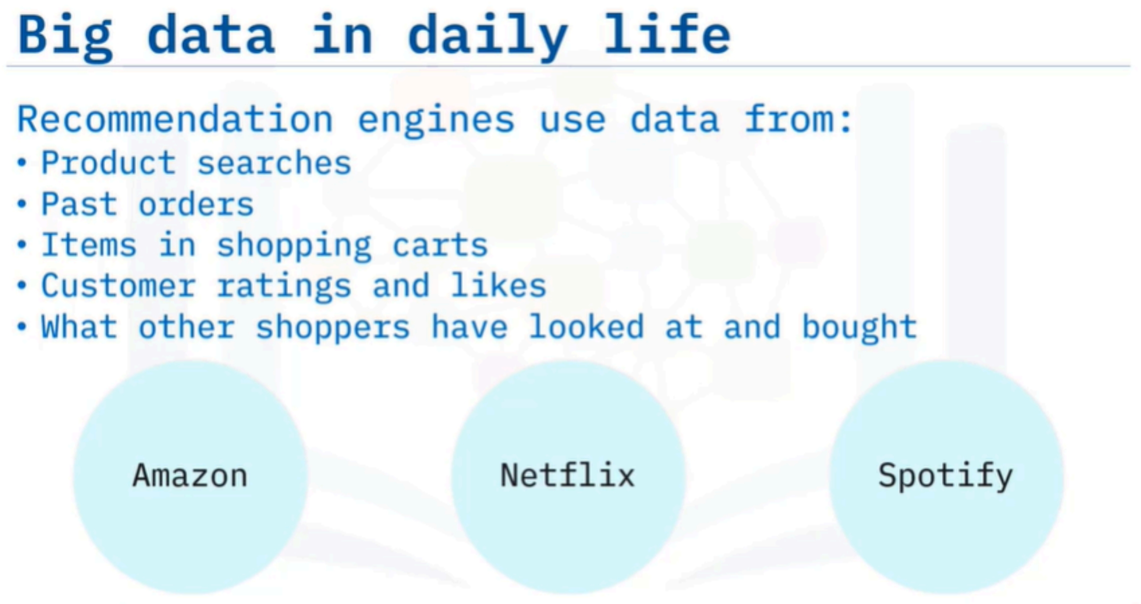
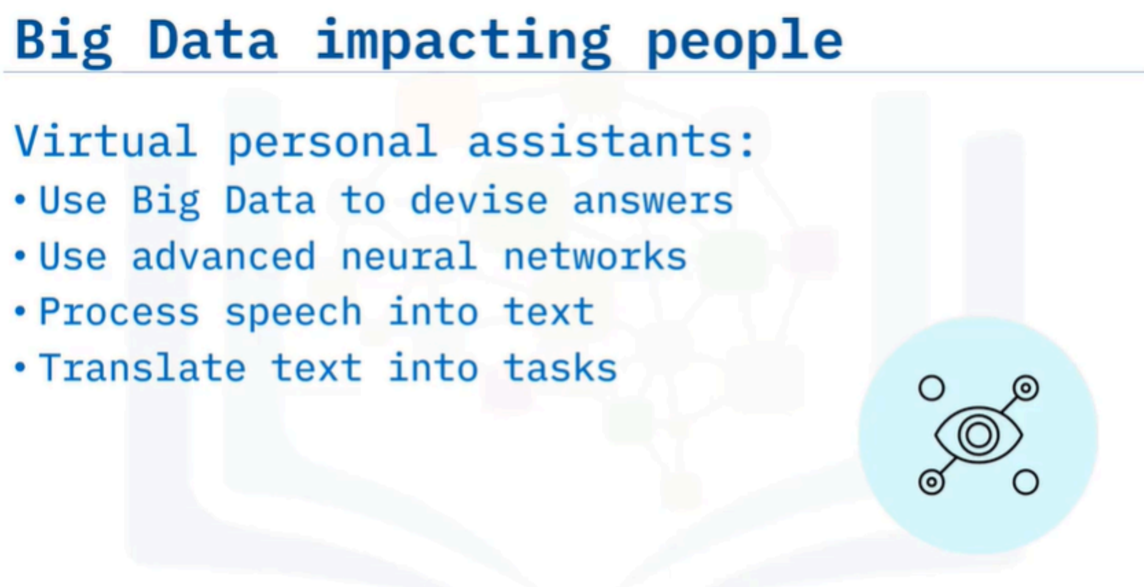
Impact of Big Data

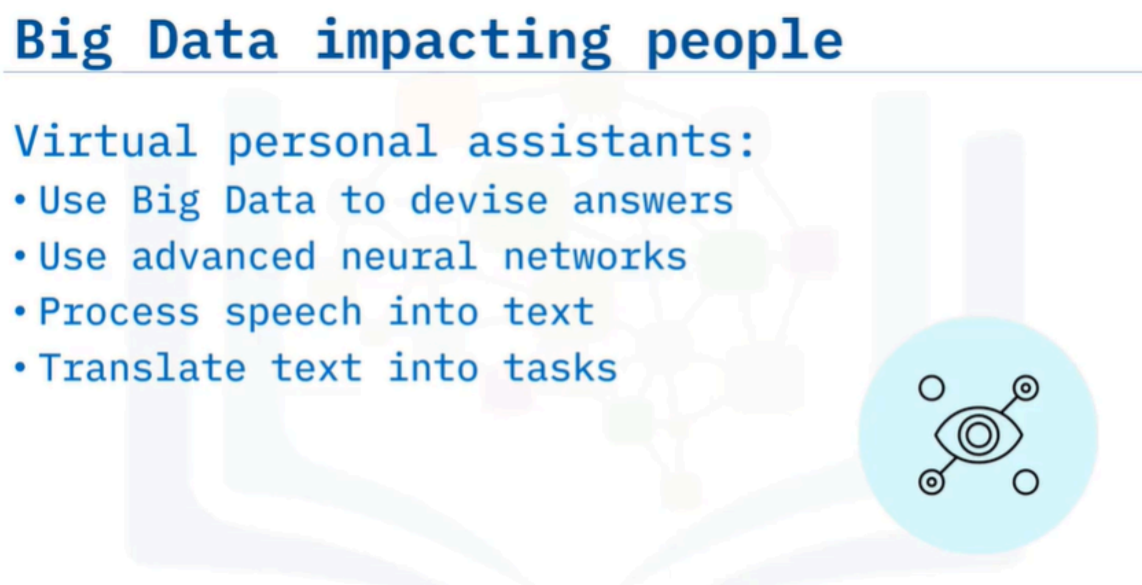
- Big Data is one of the most important subjects of this century. Yet globally, millions of people are generating and using Big Data without necessarily being aware of it. This personal data in the form of photos, videos, and text that people send to each other forms the bulk of data collected by consumer goods companies. So, it is interesting to consider how Big Data is impacting businesses and people.



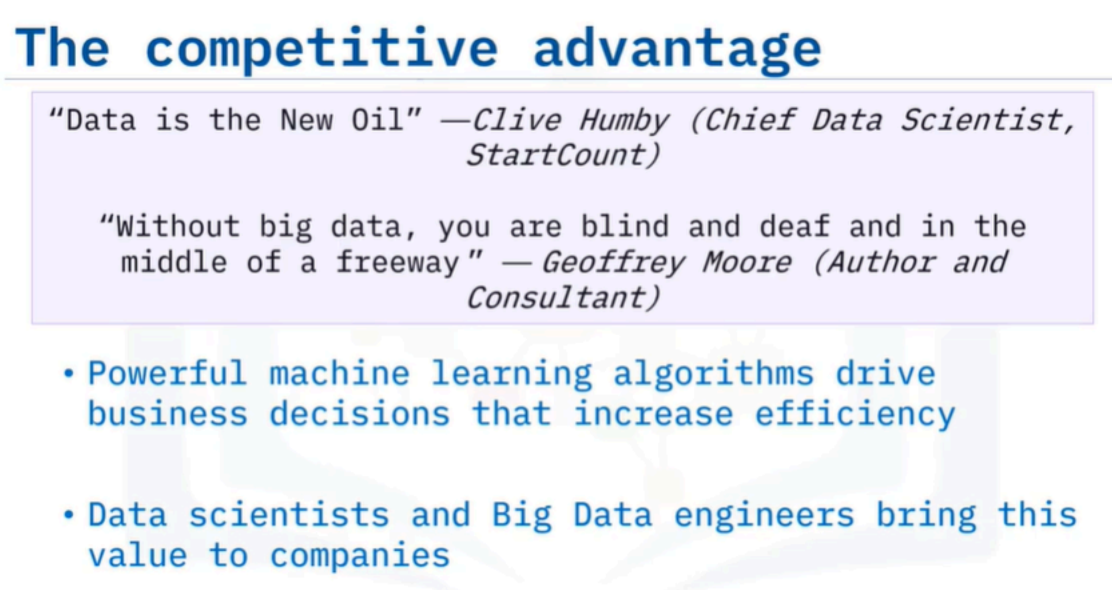
- Have you ever searched for or bought a product on Amazon? Did you notice that Amazon's recommendations are based on what the users have searched for, have bought in the past, the items they have in the virtual shopping cart, items they have rated and liked, and what other customers have viewed and purchased. Companies like Amazon, Netflix and Spotify use algorithms based on big data to make specific recommendations based on customer preferences and historical behavior. Recommendation engines are a common application of Big Data.



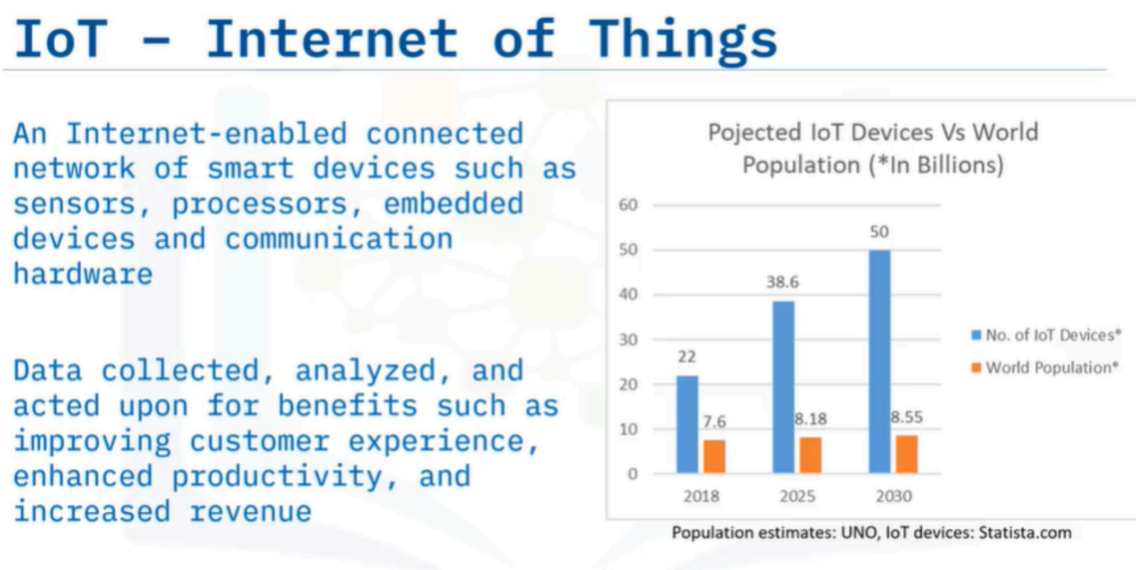
- Personal assistants like Siri on Apple devices, or Alexa on Amazon devices, use Big Data to devise answers to the infinite number of questions end-users may ask. These assistants use some of the most advanced neural networks ever made. This allows them to process speech into text and translate text into complex tasks that can be carried out across multiple applications within that device. Sources of Big Data collected for these actions include not only personal data and preferences on-device and in the cloud, but also external factors.



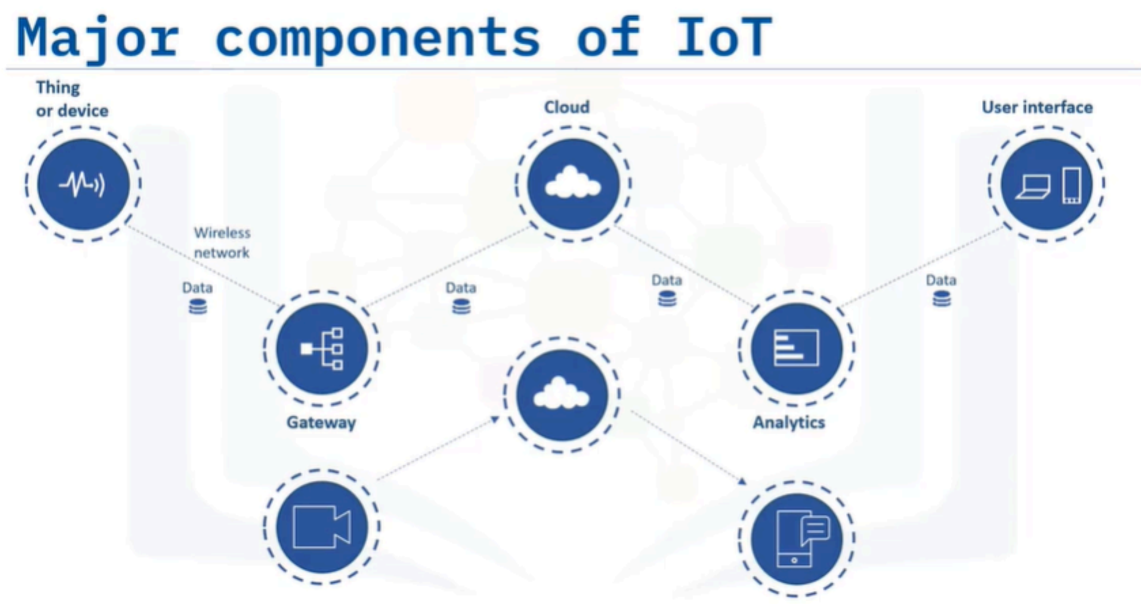
- Google Now, for example, makes recommendations before users ask for them, especially when it is linked to the users' calendars and location and sensing is enabled on the user's phone. Google Now knows where the user is, and where they need to be. It can tell users about things like traffic or the weather before they even ask for it. In this instance, the collection of Big Data is used to forecast future needs and behavior.



- Now that we have an idea of how consumers are using Big Data, let's take a look at how Big Data is impacting businesses. Clive Humby went as far as to call data the “New Oil” and Geoffrey Moore compared the value of data to our senses of sight and hearing. Big Data is fundamentally changing the way businesses compete and operate. Complex and powerful machine learning algorithms drive business decisions that ultimately increase effectiveness. In recent years the annual demand for data scientists and Big Data engineers is increasing significantly, and the Big Data skillset remains highly valuable to businesses.



- The Internet of Things (IoT) refers to a system of physical objects that are connected through the Internet. A “thing or device” can include a smart device in our homes, or a personal communication device such as a smartphone or computer. These collect and transfer massive amounts of data over the Internet without any manual intervention by using embedded technologies All of these devices gather, analyze, share, and transmit data in real time. Without Big Data, IoT devices would not hold the functionalities and capabilities which have gained them so much attention globally. Following Statista’s forecast, we will see four times more IoT devices than the total number of the entire world population by 2030.



- Let's look at major components of the IoT ecosystem. An Internet-enabled connected network of systems, that comprise smart devices and sensors, constantly produce and transmit a variety of data. This Big Data is passed through a Gateway that routes it to its destined storage in the Cloud. Data Engineers use sophisticated algorithms to make sense of this collected Big Data and compile it into an analytical form that can be interpreted by humans. The analyzed data is then used to make business and consumer decisions to enhance the customer experience and to enhance productivity. For example, a security camera could continually upload footage, and a cloud-based big data algorithm may alert the user if any activity is detected.

Summary

In this video, you learned that:

- Big Data is everywhere, and is being collected and used to drive business decisions and influence people’s lives
- Personal assistants like Siri or Alexa use Big Data to devise answers
- Google Now uses Big Data to forecast future needs and behavior
- Internet of Things or IoT devices continually generate massive volumes of data
- Big Data analytics helps companies gain insights from the data collected by IoT devices